

Review

The global burden and stigma of epilepsy

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Abstract

Epilepsy is one of the most prevalent neurological conditions and it knows no age, racial, social class, geographic, or national boundaries. The impact of epilepsy rests not only on the individual patient, but also on the family and indirectly on the community. The burden of epilepsy may be due to the physical hazards of epilepsy resulting from the unpredictability of seizures; the social exclusion as a result of negative attitudes of others toward people with epilepsy; and the stigma, as children with epilepsy may be banned from school, adults may be barred from marriage, and employment is often denied, even when seizures would not render the work unsuitable or unsafe. Furthermore, epilepsy is a disorder associated with significant psychological consequences, with increased levels of anxiety, depression, and poor self-esteem compared with people without this condition. Here we discuss some of the aspects of the global burden of epilepsy. © 2007 Elsevier Inc. All rights reserved.

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1. Introduction

Epilepsy is a neurological condition that knows no geographic, social, or racial boundaries, occurring in men and women and affecting people of all ages, though more frequently affecting young people in the first two decades of life and people over the age of 60 [1]. It has been estimated that worldwide there are at least 50 million people who have epilepsy [2,3].

It has to be acknowledged that more than 80% of people with epilepsy live in developing countries, where the condition remains largely untreated. These people could lead normal lives if properly treated, but the majority of them do not receive any effective treatment [4,5]. Despite the fact that highly effective, low-cost treatments are available, as

many as 9 of 10 people with epilepsy in Africa go untreated, as stated by Margaret Chan, the Director General of the World Health Organization (News Release WHO/4 27 January 2007). The reasons for the unavailability of treatment include: inadequate health delivery systems, lack of trained personnel, lack of essential drugs, and traditional beliefs and practices that often do not consider epilepsy as a treatable condition. This treatment gap greatly increases the burden of epilepsy and disability.

There has been a longstanding concern that, although progress continues to be made in relation to medical management of epilepsy, including the development of a number of new antiepileptic drugs (AEDs), attention to the social adjustment of individuals with the condition is still limited. Traditionally, the outcome of health conditions has been measured in terms of mortality. More recently, new indexes have been developed. These attempt to define better the need for health services and related interventions, to define health outcomes in terms of body, person, and social functioning, and to provide a common

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framework for research, clinical work, and social policy. These indexes also try to ensure the cost-effective provision and management of health care and related services, and to characterize physical, mental, social, economic, or environmental interventions that will improve the lives and levels of human functioning [3]. In this regard, the Global Disease Burden Study proposed a burden-specific measure (i.e., the disability-adjusted life year, or DALY) based on epidemiological and demographic indicators, such as prevalence and incidence rates, life expectancy, probabilities of death in different age groups, disability-adjusted life expectancy, years of life lost because of premature death, and years of living with disability. The DALY takes into account mortality and disability in the evaluation of the burden of the disease. This measure has suggested that epilepsy may contribute around 1% of the global total of days lost due to ill health. The same study estimated that the aggregate burden due to epilepsy is around 0.5% of the total disease burden. It is thus evident that epilepsy represents a considerable health problem. As stated by the ILAE/IBE/WHO Global Campaign against Epilepsy [4], the suffering and disability caused by epilepsy place a huge burden not only on people with epilepsy, but also on their families and society at large.

Epilepsy also has a considerable psychological and emotional impact. Uncontrolled seizures can be very unsettling. People may fear even going outside their homes unaccompanied. They may fear what people might think of them if they were to have a seizure in public. Across the world and throughout history, epilepsy has been a culturally devalued condition. Such devaluing often leads to people with epilepsy being stigmatized and bearing a psychosocial burden. Specific examples of the psychosocial impact of epilepsy have been reported in developed and developing countries. Lower income and quality of life compared with other people with chronic illnesses have been pointed out by a Canadian study [6]. In The Netherlands, children seem to have lower school attendance and performance [4]. In China, difficulty in finding a spouse has been described, with families of people with epilepsy feeling disgrace. In Ecuador, Ethiopia, and Kenya, social exclusion, altered relationships with spouses/parents, housing difficulties, and employment problems have been reported [4].

In this review, some of the aspects of the global burden of epilepsy are discussed.

2. The epidemiology of epilepsy

Epilepsy is among the most common serious neurological conditions, yet its epidemiology is not completely known. This is due mainly to methodological problems and to the failure to consider epilepsy's heterogeneous nature [1]. In resource-poor countries, with less sophisticated health delivery systems, case ascertainment for epilepsy is usually problematic. In these settings, there are usually also diagnostic and treatment gaps that contribute to the short-

comings in the understanding of the epidemiology of epilepsy, particularly in such settings.

2.1. Incidence

The incidence of epilepsy in developed countries is usually quoted as being between 40 and 70 per 100,000 persons per year, and it is usually higher in young children and in older people [1,7–9]. In developed countries, poorer people also seem to have a higher incidence, the reasons for which are unclear [10]. The incidence in resource-poor countries is usually much higher than in developed countries, often above 120/100,000/year. Poor sanitation, inadequate health delivery systems, and higher risk of brain infections and infestations may contribute to this [1,11], although methodological issues may also contribute.

2.2. Prevalence

The great majority of studies of the prevalence of epilepsy have reported rates between 4 and 10 per 1000 [1,8,9]. Some studies from resource-poor countries have given higher prevalence rates; these studies usually are small-scale studies from isolated geographic areas where unique genetic or environmental factors may apply or else are compounded by methodological problems [1]. Most large-scale studies in resource-poor countries report prevalence rates for active epilepsy of between 6 and 10 per 1000; many of these studies also report higher rates in rural areas.

Lifetime prevalence rates are much higher than rates for active epilepsy; this is even the case in resource-poor countries, where there is a huge treatment gap and, indeed, AEDs may not be available [1,9]. This suggests that most people developing epilepsy will either cease to have seizures or die prematurely, probably the former. Epilepsy is, however, associated with an increased mortality rate [12,13], although the impact of mortality on prevalence in resource-poor countries is not known.

2.3. Risk factors

Epilepsy is a symptom complex, and many different conditions are known to be risk factors. These vary with age and geographic location. Congenital, developmental, and genetic conditions are associated with developing epilepsy when young. Epilepsy associated with head trauma, central nervous system infections, and tumors may occur at any age, although tumors are more likely in the elderly [9]. Cerebrovascular disease is, however, the most common risk factor in the elderly [14–16]. Parasitic conditions such as falciparum malaria and neurocysticercosis are associated with epilepsy in endemic areas and are probably the most common preventable cause of epilepsy worldwide [11,17–23]. Recently, two other parasites, *Toxocara canis* and *Onchocerca volvulus*, have been suggested as important risk factors; this needs confirmation [24,25].

A family history of epilepsy seems to increase the influence of other risk factors. The susceptibility to epilepsy may be partly genetically determined, and this may interact with brain maturation and environmental factors. These interactions may be responsible for the shortcomings of our understanding of the dynamics of epilepsy in the population [9,11]. For instance, the relative risk of developing epilepsy with different conditions in different populations is not known.

2.4. Prognosis for remission

In developed countries, more than 70% of patients on optimum treatment achieve long-term remission, usually within 5 years of diagnosis; the prospect of remission decreases as time elapses [1]. Predictors of a good outcome include age at onset, number of early seizures [26,27], early response to AEDs [28], and some specific EEG findings. In any individual patient, outcome and response to treatment may be inherent, and seizure control in some patients may be difficult from the onset [1,11,29].

AEDs are highly successful at suppressing seizures, but their influence on prognosis remains unknown [29]. In the developed world, the overall good prognosis is often attributed to the widespread use of AEDs. In resource-poor countries, however, despite the lack of AEDs, many patients enter long-term remission, supporting the suggestion that prognosis is inherent and dependent on the cause of the epilepsy [1,11].

2.5. Prognosis for life

People with epilepsy have increased morbidity. This may be physical (e.g., fractures, scalding, bruising), occurring directly as result of seizures. There is also comorbidity with other systemic diseases [30] and increased psychosocial problems [31]. People with epilepsy, particularly if chronic, have an increased risk of premature death [11,32]. Symptomatic epilepsy may reduce life expectancy by up to 18 years [31]. Sudden death, trauma, suicide, pneumonia, and status epilepticus have been identified as being more common in people with epilepsy [33]. Little is known about mortality in resource-poor countries, as no proper studies addressing this issue have been carried out. Circumstantial evidence, however, suggests that the rate is higher, helping to explain the discrepancy between the higher incidence and relatively low prevalence of active epilepsy in these countries [1].

3. Comorbid conditions

About 40% of individuals with epilepsy between the ages of 4 and 15 have one or more additional neurological disorders. The most common ones are cognitive delays, speech/language disabilities, and other specific learning disabilities. Up to a fourth of patients with epilepsy have learning disabilities, and, conversely, up to half of all patients with learning disabilities have seizure disorders

[34]. Both conditions share a certain degree of stigmatization and a common heritage of heterogeneity [35]. In addition, learning-disabled persons with epilepsy, particularly those with more severe disabilities, have greater health care needs than many other people. They may have a number of health problems and often have also a physical disability, such as paralysis, hemiplegia, and hemiparesis, often with feeding and swallowing difficulties, leading to difficulties in taking AEDs and other medications; poor nutrition, which can worsen epilepsy; and ailments such as constipation and urinary tract infections, which may lead to exacerbation of seizures. It is often difficult to identify these infections, and the first sign may be an increase in seizures. Additionally, behavioral problems often occur. Where these health problems are not dealt with, they can significantly affect quality of life of the person and represent a significant burden to the individual as well as those in the immediate surroundings. This is compounded in resource-poor countries where incipient health and care delivery systems do not have the resources to deliver any amount of care to most of these people.

4. Cognitive functioning

Apart from severe learning disabilities, epilepsy might impact on learning in several ways. One major area of cognitive malfunctioning in people with epilepsy is memory impairment. This impairment can range from poor concentration and minor forgetfulness to gross clouding of consciousness and disorientation. Seizures can also affect learning by reducing alertness and by interfering with short-term information storage and abstraction. Frequent and uncontrolled seizures impair learning of new information because of the amount of time the person is unaware of the environment. Nighttime seizures can disrupt the consolidation of memory and affect language functions. In children, all these issues are of great relevance because, whenever cognitive impairments occur, even if these are temporary, they may affect educational progress and lead to restricted occupational opportunities in later life [36].

Cognitive impairments can also be a side effect of the various AEDs used. AEDs have been associated with learning difficulties, behavior changes, and memory impairment. The most commonly implicated drug is phenobarbital, which may cause hyperactivity and memory impairment. This is the AED most often available in resource-poor countries, where there is a general scarcity of medicines including AEDs. This may create some difficulties, although an individual benefit–risk assessment should be carried out in each individual case and this will usually be in favor of treatment.

5. Psychiatric and behavioral problems

Epilepsy is a chronic condition associated with increased social discrimination. People with intractable seizures are at constant risk of becoming unconscious and of falling and sustaining injuries and, in public, social embarrassment.

Table 1

Rates of prevalence of some psychiatric disorders in patients with epilepsy as compared with those reported in the general population

	Patients with epilepsy	General population
Depression	11–60%	Dysthymia 3.3% Major depression 4.9–17%
Psychosis	2–9%	Schizophrenia 1% Schizophreniform disorder 0.2%
ADHD	12–37%	4–12%
Generalized anxiety disorders	15–25%	5–7%
Social phobia	15–20%	10–12%
Panic disorder	5–21%	0.5–3%

As with any chronic condition, epilepsy can be linked to demoralization and a negative perspective on life and, further, to disturbances of affect and mood [37]. Psychiatric and psychological disorders are underrecognized and undertreated in people with epilepsy. The result is that epilepsy is complicated by the presence of psychiatric disturbances and behavioral problems that often take a greater toll on the quality of life than the seizures.

Depression, anxiety, psychoses, and attention deficit disorders are common among people with epilepsy. Table 1 compares the prevalence rates of these psychiatric disorders between people with epilepsy and the general population. The higher prevalence rates in people with epilepsy, when compared with the general population, further complicate the burden of epilepsy; disorders such as mental health problems often are as highly stigmatized as epilepsy, particularly in settings with less sophisticated health and care delivery systems.

6. Stigma

People with disabilities are among the most vulnerable in any society. This vulnerability is even greater among those with hidden disabilities such as epilepsy and other neurological conditions and intellectual disabilities. Although the vulnerability of people living with epilepsy may be partly attributed to the disorder itself, “all chronic medical conditions have an impact on daily life, but the impact of epilepsy is greater”; the particular stigma associated with epilepsy brings a susceptibility of its own. Stigmatization leads to discrimination, and people with epilepsy have been the target of prejudicial behavior in many spheres of life, over many centuries and in many cultures [38]. As was once stated “The history of epilepsy can be summarised as 4000 years of ignorance, superstition and stigma, followed by 100 years of knowledge, superstition and stigma” [39].

Understanding and dispelling stigma is one of the stated aims of the Global Campaign Against Epilepsy (GCAE) “Out of the Shadows” initiative [4,40]. Though the implications of epilepsy-related stigma for disease burden and quality of life are difficult to calculate [40], they are almost certainly of great significance in resource-poor countries. In these countries, epilepsy represents a major public health problem characterized by lack of prioritization and infrastructural support and a “treatment gap”. Next to clinical interventions, interventions are also required that recognize

epilepsy’s social course and the relevance and the importance of epilepsy-related stigma to overall disease burden and impact on quality of life [41].

Discrimination against people with epilepsy in the workplace and with respect to access to education is not uncommon for many people affected by the condition. Violations of human rights are often more subtle and include social ostracism, being overlooked for promotion at work, and denial of the right to participate in many of the social activities taken for granted by others in the community.

Civil and human rights violations are more evident in resource-poor countries, especially where there has been a history of generalized rights abuse. However, such practices are not limited to developing countries and traditional societies, and there is evidence of rights violations in developed countries and societies that are regarded as having impeccable human rights records in other respects.

Failure to secure health insurance protection leaves people with epilepsy unnecessarily exposed to risks that largely have no bearing on their epilepsy at all, and restrictions on obtaining insurance are not limited to developing countries. It has been reported that as many as 36% of people with epilepsy were refused one or more types of insurance in the United Kingdom [42]. This discriminatory behavior translates into a restriction of the right to make adequate financial provision in guarding against risks to the individual and his or her family. It has been reported that people with developmental disabilities, including epilepsy, experience a heightened risk of becoming victims of violence and abuse [43]. Research into the vocational interests of people with epilepsy shows that people with epilepsy often are not encouraged to follow their training of choice because of the suspected consequences of having epilepsy [40]. From reading the accounts, it becomes clear that although specific practices may vary from country to country, discriminatory attitudes and prejudicial behavior toward people with epilepsy are common across the globe.

There are two ways to reduce the stigma of epilepsy, which is a significant part of the psychosocial burden of epilepsy worldwide. The first relates to people with epilepsy themselves, who need support to counter prevailing negative stereotypes and reduce their experience of stigma. Epilepsy associations are moving rapidly from the provision of support and information to an increasingly political role, agitating for better services and less discrimination for their members [44]. At the same time, campaigns such as the Global Campaign Against Epilepsy (2001) are raising the

profile of the disorder with governments and their health system planners and providers. The second way is to support people with epilepsy individually to develop resourcefulness and resilience in putting the stigma of their condition aside. A critical factor here is increased knowledge, as the lack of it can increase a sense of stigma on the part of affected persons [45,46]. Targeted educational programs and counseling for people with epilepsy and their families are therefore clearly indicated.

Initiatives are also required that focus on changing negative public attitudes. A number of strategies have been proposed, including providing education and information, advocacy, and increasing the level of contact between people with epilepsy and people without epilepsy. The success of a number of initiatives suggests that much can be done to reduce the experience of stigma among people with epilepsy and, so, lessen the burden of their condition (Epilepsy in the WHO European Region).

7. Legislation

Legislation based on centuries of stigmatization has existed until recently or still does in many countries. *Sakikku* [All Diseases], a text on medicine written in 1700 B.C., during the reign of King Hammurabi in Babylon, describes a criminal and commercial law enabling slave owners to return a slave who had a seizure within a month of purchase and to be refunded the entire sum paid.

In many countries, laws impacting the lives of people with epilepsy are outdated, failing to adequately promote and protect their human rights and, in some cases, even violating these rights. Antimarrriage laws, with penalties for infringement, were introduced in various countries more than 200 years ago. They have remained, but have not been implemented for many years. In the United States, for instance, until 1956, 17 states prohibited people with epilepsy from marrying. The last state to repeal its law forbidding marriage to people with epilepsy did so only in 1980. In the United Kingdom, a law prohibiting people with epilepsy from marrying was repealed in 1970. In some parts of the world, epilepsy is still commonly viewed as a reason for annulment or prohibition of marriage [40].

Unemployment and underemployment exist worldwide. Employment and legal devaluations have further authenticated the stigma attached to epilepsy in modern Western, Eastern, and Southern cultures [40].

The WHO reports on Epilepsy in the African Region [38] and the Western Pacific Region [38,47] provide numerous accounts of civil and human rights violations against people with epilepsy across these regions, which comprise mainly developing countries. However, examples of legislation based on centuries of stigmatization can be found in many countries all over the world. And yet, in other countries, there is no legislation in this area [40].

The restrictions on full participation in community life can marginalize people with epilepsy from society. For example, ineligibility for a driving license frequently

imposes restrictions on social participation and choice of employment. Discrimination in access to education is not unusual for people affected by the condition [48].

Although specific practices vary from country to country, discriminatory attitudes and prejudicial behavior toward people with epilepsy are common around the globe. Legislation is an important means of addressing these problems and challenges. Well-crafted legislation based on internationally accepted human rights standards can prevent violations and discrimination, promote and protect human rights, enhance the autonomy and liberty of people with epilepsy, and improve equity in access to health care services and community integration. Legislation can serve to legally enforce the goals and objectives of policies and programs related to epilepsy.

The right to equal treatment is clearly described in law (United Nations and European Union Treaty of Amsterdam, 1997) and deserves the protection of enforceable legal remedies. Accordingly, strategies aimed at promoting the social and professional integration of people with epilepsy and at rooting out the prejudice they encounter should be secured and underpinned by equal opportunity legislation. Such legislation should render it illegal to treat people with epilepsy less favorably than others, unless clear justification for doing so can be proven. Blanket restrictions and barriers automatically generated in response to epilepsy should be prohibited. The principle that people are entitled to individual assessment of their abilities and the risk they present to others should be endorsed by law [49].

The current reality, however, is that the laws that affect the lives of people with epilepsy are often outdated. They frequently fail to promote and protect human rights adequately, and they sometimes actively promote the violation of rights. In many countries, there is no protective legislation, and there are ample examples of legislation based on centuries of stigmatization.

8. The economic burden of epilepsy

Epilepsy contributed more than 7 million DALY (0.5%) to the global burden of disease in 2000 [3,38]. It is apparent that close to 90% of the worldwide burden of epilepsy is to be found in developing regions, with more than half occurring in the 39% of the global population living in countries with the highest levels of premature mortality (and lowest levels of income). An age gradient is also apparent, with the vast majority of epilepsy-related deaths and disability in childhood and adolescence occurring in developing regions; later in the life cycle, the proportion decreases because of the relatively greater survival rates into older age of people living in more economically developed regions [47].

Economic assessments of the national burden of epilepsy have been conducted in a number of high-income countries [50,51] and more recently in India [52], and have clearly demonstrated the significant economic implications the disorder has in terms of health care service needs,

premature mortality, and lost work productivity. For example, the Indian study calculated that the total cost per case of these consequences of epilepsy amounted to US \$344 per year (equivalent to 88% of average income per capita), and that the total cost for the estimated 5 million cases resident in India was equivalent to 0.5% of its gross national product. As such studies differ with respect to the exact methods used, as well as underlying cost structures within the health system, they are currently of most use at the level of individual countries, where they can serve to draw attention to the wide-ranging resource implications and needs of people with epilepsy [47].

9. Conclusions

Despite the lack of specific data on a number of areas, there is little doubt that epilepsy imposes a substantial burden on individuals affected by it, their families, the community, and society as a whole. This increased burden is represented in many domains including health, psychosocial well-being, and economic status. This burden is even more of an issue in resource-poor countries where treatment and care for people with epilepsy are usually not available. More needs to be done to address these issues and, eventually, to reduce the burden of epilepsy worldwide.

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